



**SİVAS
BİLİM VE TEKNOLOJİ
ÜNİVERSİTESİ**

Water Level Monitoring and Pump Controller

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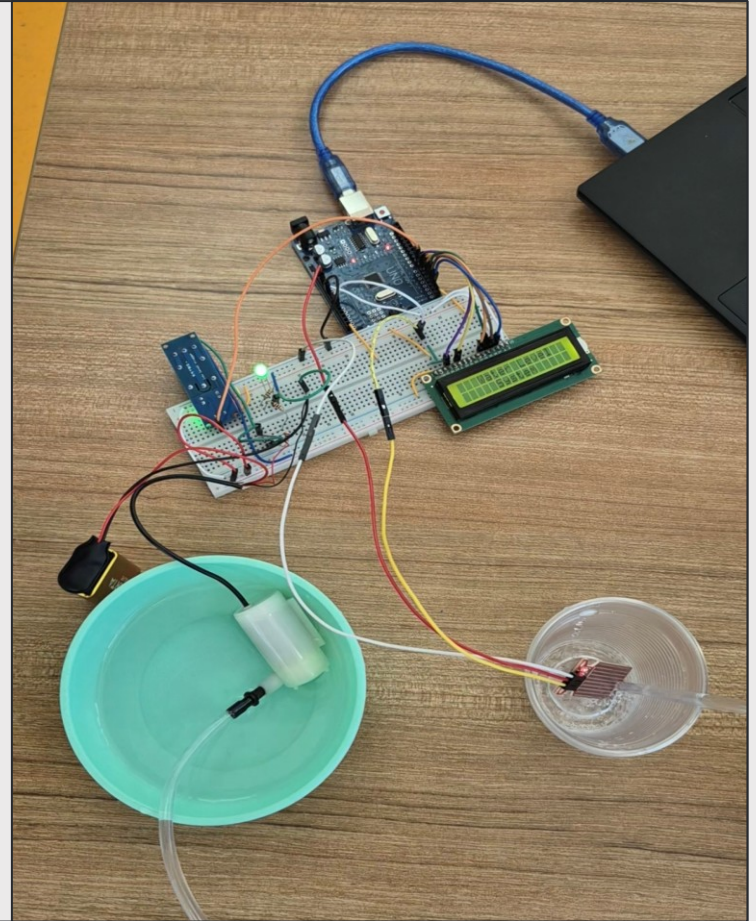
Circuit Design and Working

Visual and functional breakdown of the circuit both in Proteus and real-life setup

01

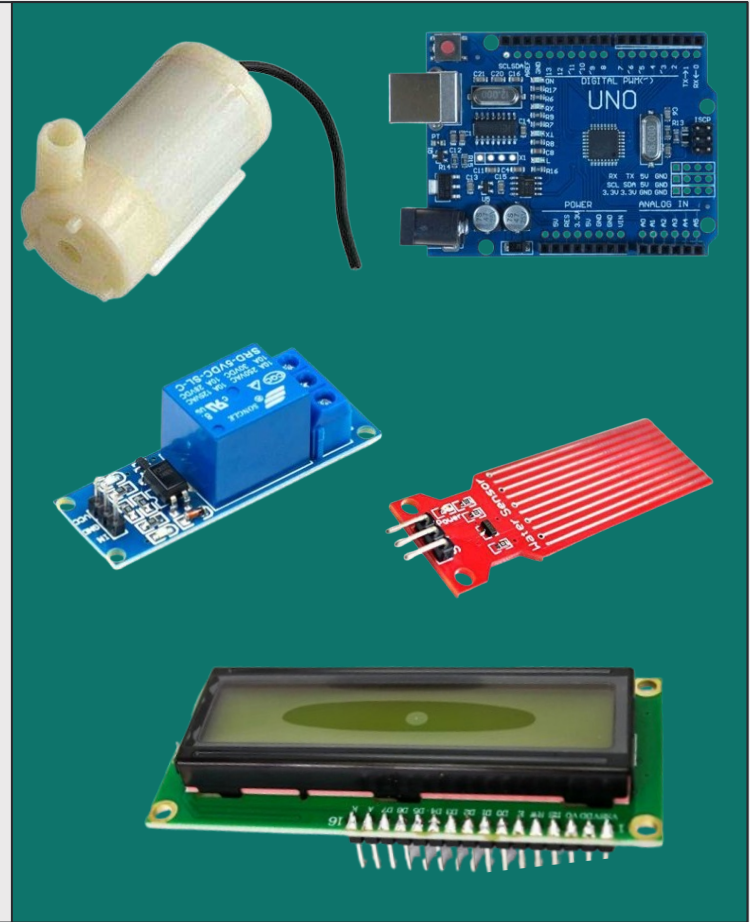
Introducton

- What is a Water Level Monitoring and Pump Controller?
- Purpose of project.



02

Hardware Components



Hardware Components



Arduino UNO R3

It acts as the brain of the entire system



Water Level Sensor

This sensor is used to detect the water level in a tank



LCD

To display textual information to the user



Water Pump

DC water pump to transfer water

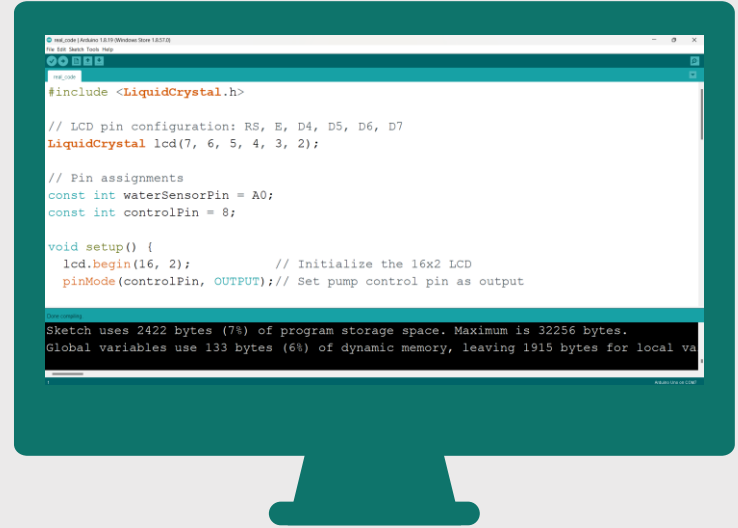


Relay Module

The relay module is as switch to control the pump

03

Code Explanation



Code Explanation

LCD Setup

```
#include <LiquidCrystal.h>

// LCD pin configuration: RS, E, D4, D5, D6, D7
LiquidCrystal lcd(7, 6, 5, 4, 3, 2);
```

Pin Assignments

```
// Pin assignments
const int waterSensorPin = A0;
const int controlPin = 8;
```

Code Explanation

Setup Function

```
void setup() {  
  lcd.begin(16, 2);           // Initialize the 16x2 LCD  
  pinMode(controlPin, OUTPUT); // Set pump control pin as output  
  
  // Display welcome message  
  lcd.setCursor(3, 0);  
  lcd.print("Water Level");  
  lcd.setCursor(5, 1);  
  lcd.print("System");  
  delay(2000);                // Wait for 2 seconds  
  lcd.clear();                // Clear LCD  
}
```


Code Explanation

Main Loop Function

```
void loop() {  
  int sensorValue = analogRead(waterSensorPin);  
  sensorValue = constrain(sensorValue, 10, 250);  
  int levelPercent = map(sensorValue, 10, 250, 0, 100);
```

Printing “Level” on LCD

```
  lcd.setCursor(0, 0);  
  lcd.print("Level: ");
```

Code Explanation

Water Level Status

```
// Display level description
if (levelPercent >= 90) {
    lcd.print("HIGH  ");
} else if (levelPercent >= 75) {
    lcd.print("HALF  ");
} else if (levelPercent >= 25) {
    lcd.print("LOW   ");
} else {
    lcd.print("EMPTY ");
}
```

Code Explanation

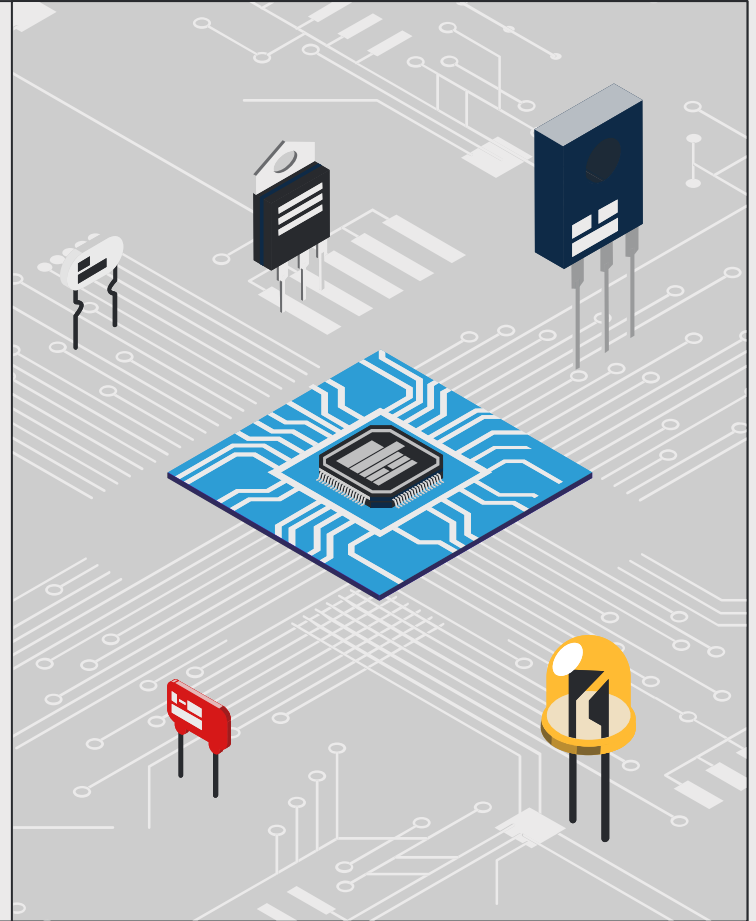
Pump Control Logic

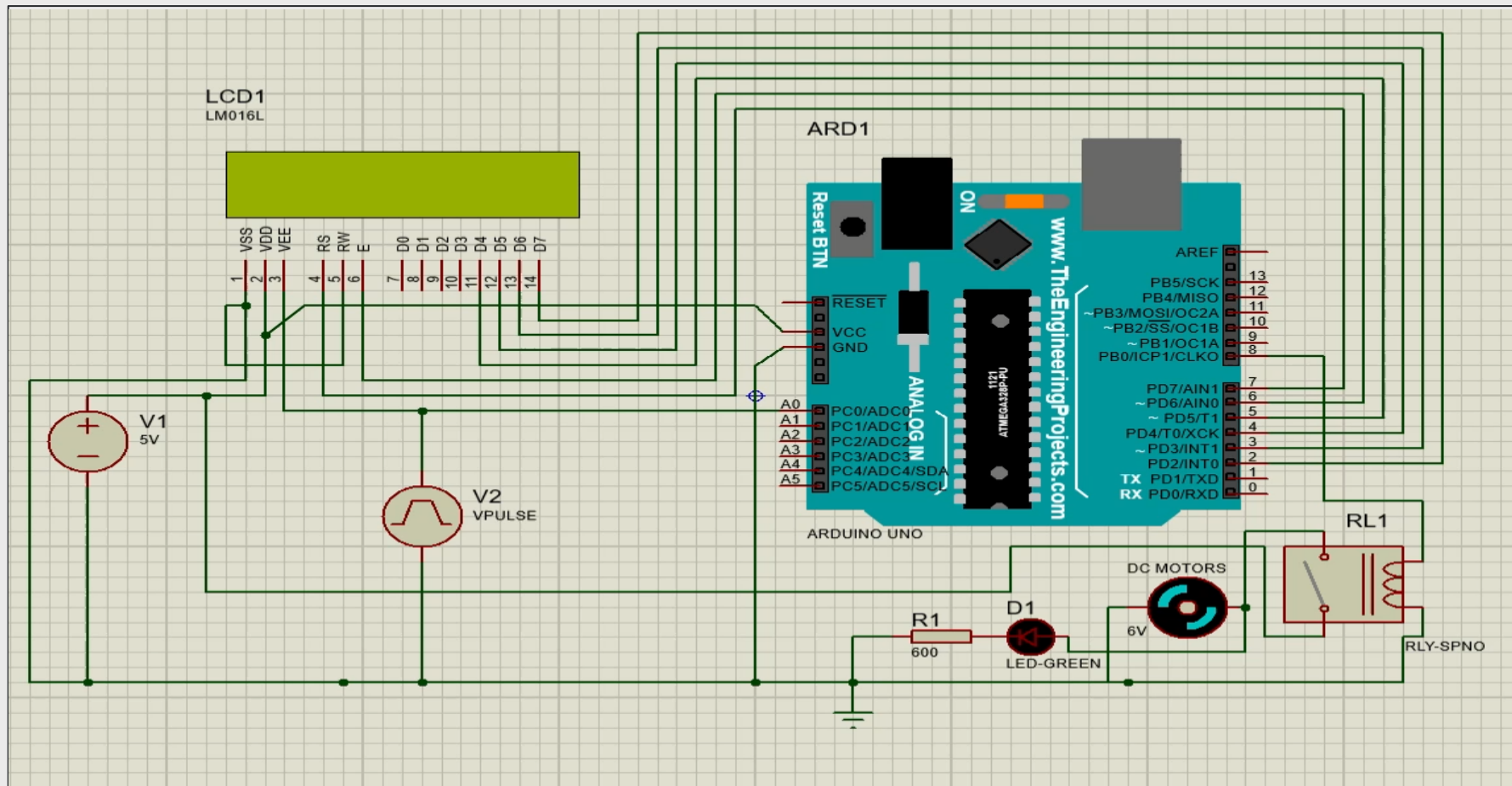
```
// Pump logic: ON until FULL
if (levelPercent >= 95) {
    digitalWrite(controlPin, HIGH);    // Turn pump OFF
    lcd.setCursor(0, 1);
    lcd.print("Pump: OFF    ");
} else {
    digitalWrite(controlPin, LOW);     // Turn pump ON
    lcd.setCursor(0, 1);
    lcd.print("Pump: ON     ");
}

delay(500);
}
```

04

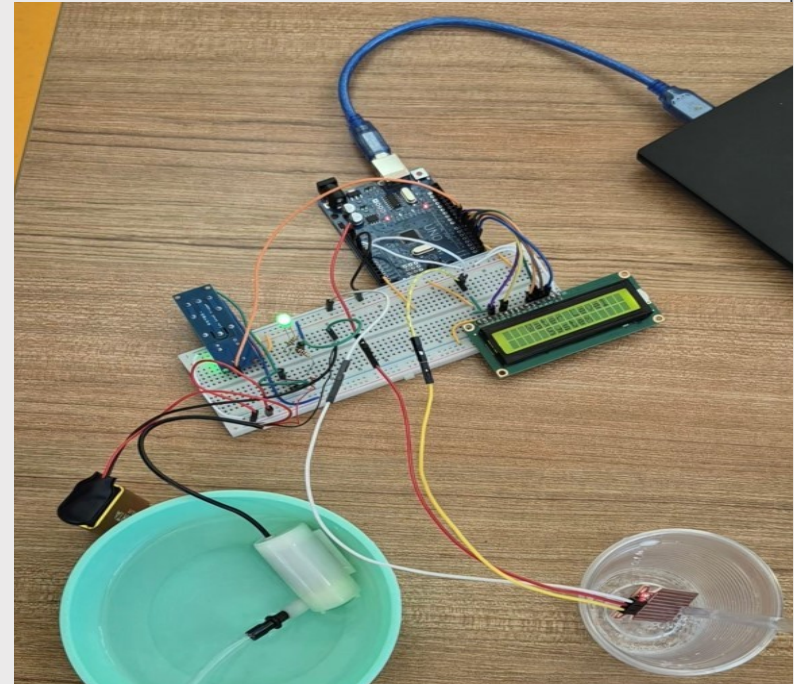
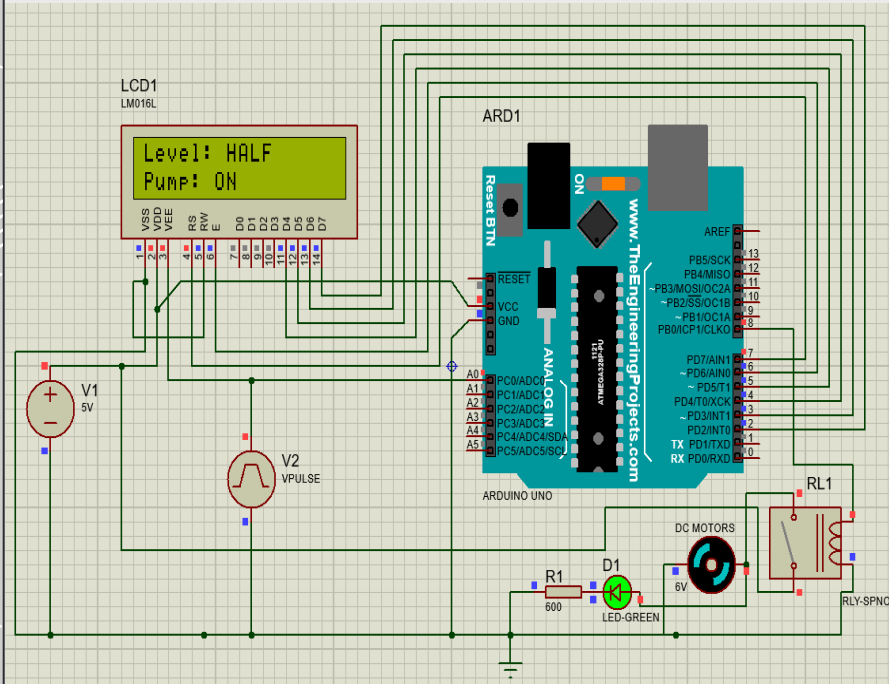
Circuit Design and Working

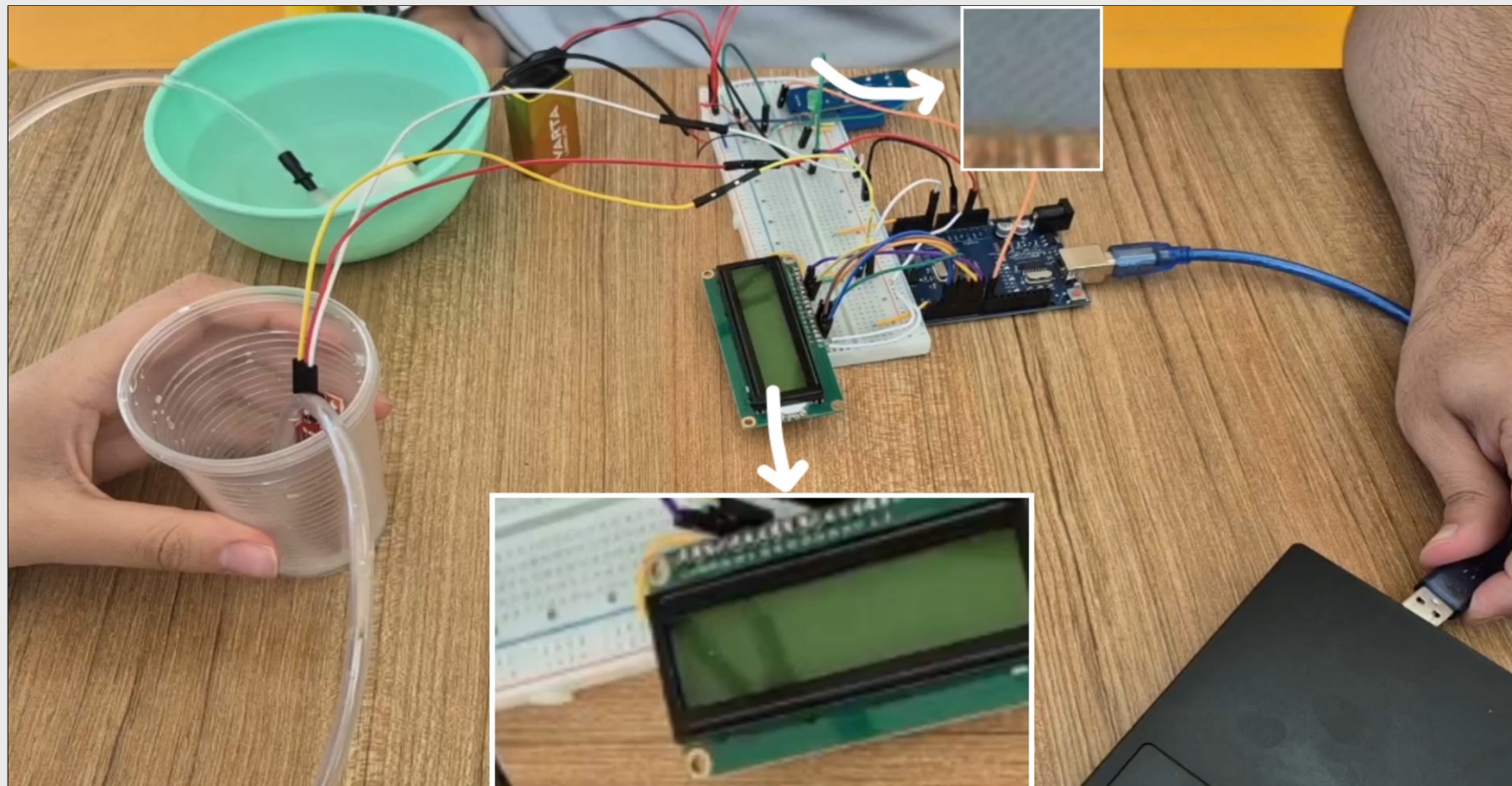




Circuit Design and Working

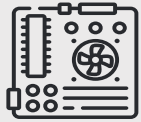
Differences Between simulation and Real-Life circuit



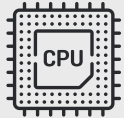




THANKS!



Circuit Theory I



Programming

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Physics II



Calculus II